



AMD DEVELOPER HACKATHON: ACT II — TRACK 3, UNICORN

AegisMed

A virtual board of AI specialist physicians that helps doctors get second opinions on complex and undifferentiated cases.

Google Gemma

AMD Instinct MI300X

Fireworks AI

AMD Developer Cloud

THE PROBLEM

Patients wait 5–7 years for a correct rare-disease diagnosis

Diagnostic reasoning is distributed across specialties. No single physician can hold all the knowledge needed to confidently work up a complex case — the value of a multidisciplinary case conference applies to any undifferentiated presentation, not only rare disease.

5–7

years, average time to a correct rare-disease diagnosis

7+

specialties often needed to see the full picture of one case

1

physician, in practice, holding the case at any given moment

THE IDEA

Recreate the multidisciplinary case conference — convened in 60 seconds

The gold-standard way hospitals actually solve hard cases is a structured review by specialists across domains. AegisMed rebuilds that as software: seven AI specialist agents instead of seven busy humans, available on demand.

1

Independent opinions

Each specialist reasons separately — real second opinions can't anchor on each other's first guess.

2

Grounded in real evidence

Citations are attached from a knowledge base, never recalled from the model's memory.

3

Token-efficient by design

Smart routing convenes only the 3–4 specialists a case actually needs, not all 7 every time.

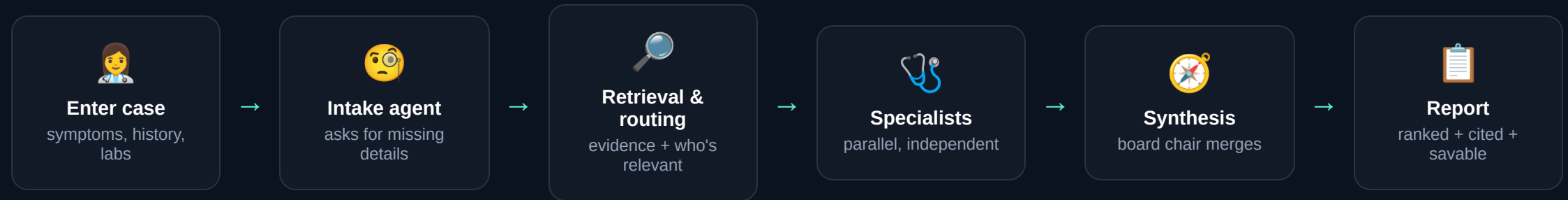
4

Decision support, not autonomy

Every output must be verified by a licensed physician — the disclaimer is enforced throughout.

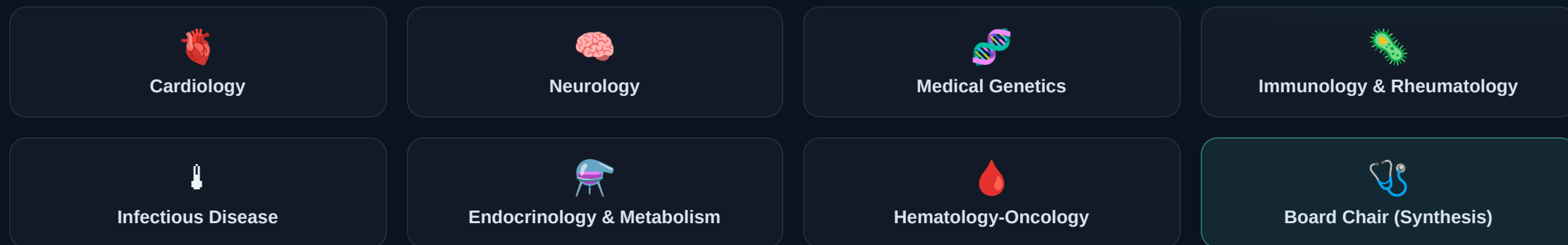
HOW IT WORKS

One case, six clinical steps



Output: a ranked differential diagnosis with rare-disease flags, points of agreement/disagreement, the single most valuable next test, immediate safety actions, and a do-not-miss warning — all within a bounded ~28-second response budget.

Seven specialists, one board chair



Each "agent" is the same Gemma model given a different system-prompt role — one model, seven personas, running **in parallel** via `asyncio.gather`, so seven opinions take the time of one. Any specialist may say "nothing in my domain" rather than invent a diagnosis — a confident non-answer is treated as genuinely useful.

Every inference call runs on AMD

Path A — Default

Managed inference on AMD Instinct

Fireworks AI serves Google Gemma on **AMD Instinct MI300X** GPUs. Zero setup — every specialist and synthesis call already runs on AMD hardware the moment a Fireworks key is set.

Path B — Self-hosted

Gemma on your own AMD Developer Cloud GPU

Set `LLM_BASE_URL` to point AegisMed at Gemma served by vLLM/Ollama on an **AMD Developer Cloud** GPU instance you provision — app and model both run on AMD infrastructure, no third-party inference service in the loop.

Also on AMD Developer Cloud

The AegisMed app itself is deployed with `deploy/amd-cloud.sh` — a one-command script that builds and runs the Docker image on an AMD Developer Cloud instance, giving judges a live, publicly reachable demo URL. `GET /health` reports the active endpoint so the AMD path in use is verifiable at a glance.

The AI is never the source of a citation

Asking an LLM to "cite sources" produces confident, plausible, and often completely fabricated references — in medicine a fabricated citation is worse than none. AegisMed follows one rule throughout the codebase: citations are always attached deterministically, never recalled from the model.

Verified references

Orphanet, OMIM, PubMed, and GARD links resolved from a ~10,700-disease knowledge base — real URLs, every time.

Guideline search links

Live, query-filtered links to PubMed, Cochrane, NICE, TRIP, MedlinePlus, NCBI Bookshelf, and Guideline Central — reordered per practice region (US/UK/EU).

Evidence-first retrieval

A retrieval agent proposes what to look up before diagnosing; the specialists cite those verified leads by name, never inventing papers or PMIDs.

Scored against real, public rare-disease cases

AegisMed is evaluated on held-out cases from **RareBench** and **CUPCase** (both Apache-2.0), mixing coded symptom lists and narrative doctor write-ups — not synthetic or hand-picked examples.

The two-step harness

- `python data/build_dataset.py` — downloads & converts public cases (free, no key)
- `python eval/run_eval.py` — runs the full board on each case, checks whether the known diagnosis (or a synonym) appears in the output
- Mirrors the real app flow: intake questions are auto-answered from the case, then the board runs

The headline metric

"The correct diagnosis was surfaced in **X%** of held-out cases" — written to `eval/results.md`, broken down per data source.

Requires a real `FIREWORKS_API_KEY` to be meaningful — demo mode returns fixed sample output, so the harness warns explicitly when scores wouldn't be real.

From solo clinician to regulated deployment

Who it's for today

- ✓ Solo clinicians — free second opinions on hard cases
- ✓ Medical schools — teaching mode grades vs. board
- ✓ Hospitals — case history & team collaboration (B2B)
- ✓ Telehealth platforms — API integration mid-consult

What's next ([docs/ROADMAP.md](#))

- ✓ Geographic expansion — region-aware guidelines (in progress)
- ✓ FHIR/EHR read integration for auto-populated cases
- ✓ FDA 510(k) / SaMD regulatory pathway
- ✓ Outcome tracking & specialist-prompt feedback loop

HONEST ASSESSMENT

Strong prototype today — a clear path to production

WORKS TODAY

One-command Docker, demo & real-AI modes, 48 passing tests

Bounded ≤28s response time, clean timeout handling

Verified, never-invented citations throughout

Graceful fallback if any data file is missing

PRODUCTION GAP

File-based case store

No auth / access control

No PHI/HIPAA compliance

No regulatory clearance

WHAT CLOSES IT

Postgres with migrations & pooling

OAuth2/SSO, per-case permissions, audit trail

Encryption at rest/in transit, signed BAAs

FDA 510(k) / SaMD pathway (Phase 4a)

Verdict: AegisMed demonstrates the multi-agent board concept end-to-end with real fundamentals — not yet deployable for real patient care, and every gap above is documented with what closing it would take. See **docs/PRODUCTION_READINESS.md** for the full analysis.

TRY IT

See the board convene

Zero-cost demo mode, real AI with a Fireworks or self-hosted-on-AMD key, or read the code — everything is public and documented.

GITHUB

github.com/wachirawut2023/AegisMed

LIVE DEMO

deployed via `deploy/amd-cloud.sh` on AMD Developer Cloud

QUICKSTART

`docker compose up --build → localhost:8000`

FULL WALKTHROUGH

`docs/PROJECT_SUMMARY.md`

AegisMed is a clinical decision-support prototype for licensed physicians. It does not provide medical advice, diagnosis, or treatment. All output must be verified by a qualified clinician. Built for the AMD Developer Hackathon: ACT II, Track 3 (Unicorn).